



Exploring Drawbridge Transmon Coherence

Zlatko Minev

Kyle Serniak
Ioan Pop
Zaki Leghtas
Uri Vool

Gianluigi Catelani
Reinier Heeres
Chen Wang
Nissim Ofek

Katrina Sliwa
Teresa Brecht
Yiwen Chu

Thanks to:

Phil, Mike, Brian, Max, Chris, Matt, Shyam,
Steven, Yvonne, Qulab, RSL, and YINQE

Luigi Frunzio, Rob Schoelkopf, Michel Devoret

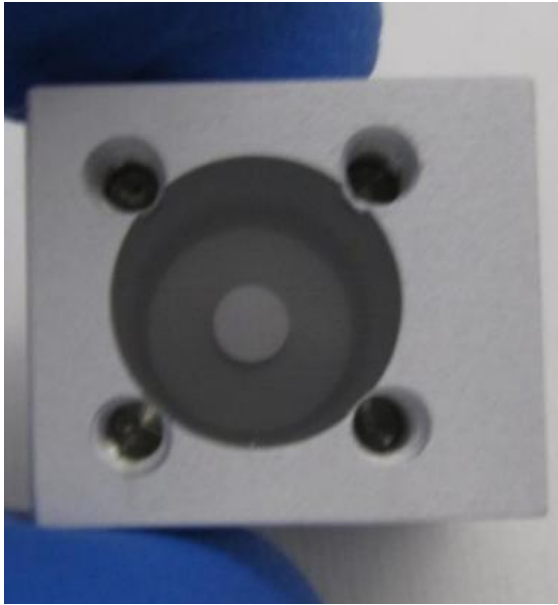


I A R P A
BE THE FUTURE

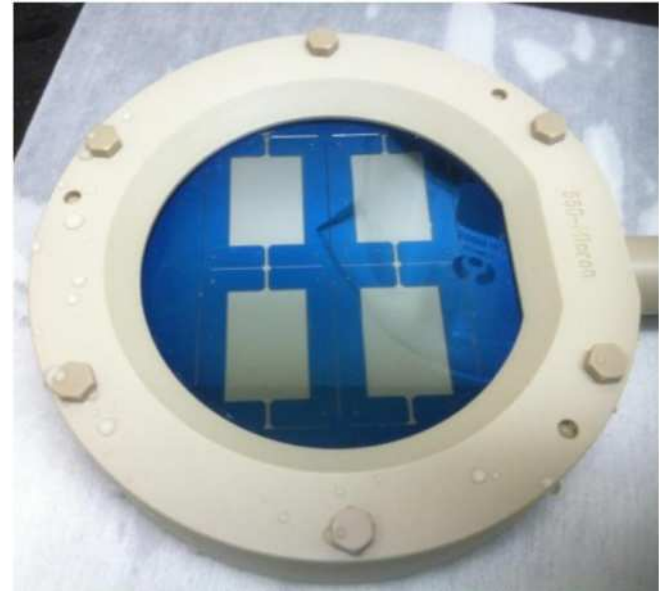
Outline

- I. What is the 2.5D multilayer architecture?
- II. Proof of concept devices
- III. System parameters & photon number splitting regime
- IV. Coherence through dynamical decoupling
- V. Quasiparticle dynamics

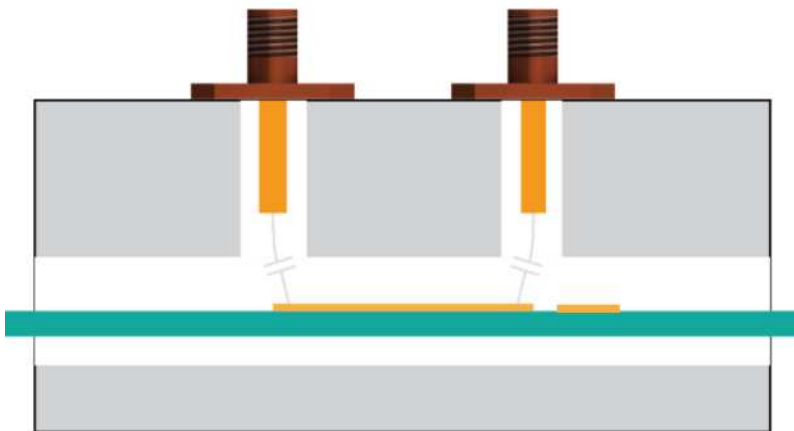
Approaches to Combining Coherence and Complexity



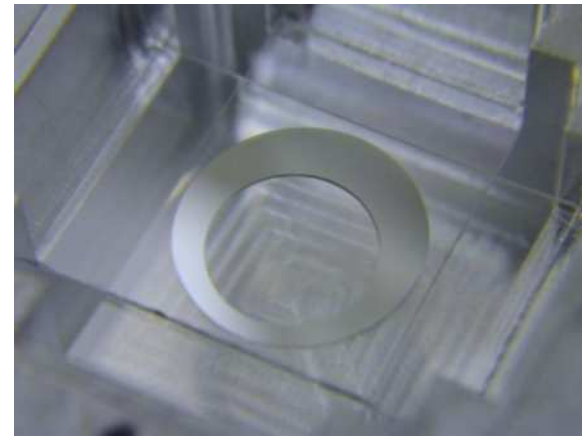
Coax cavities (MLS 9/12, 2/14)



Micro-machined cavities (MLS 9/12, 12/13, 1/15)

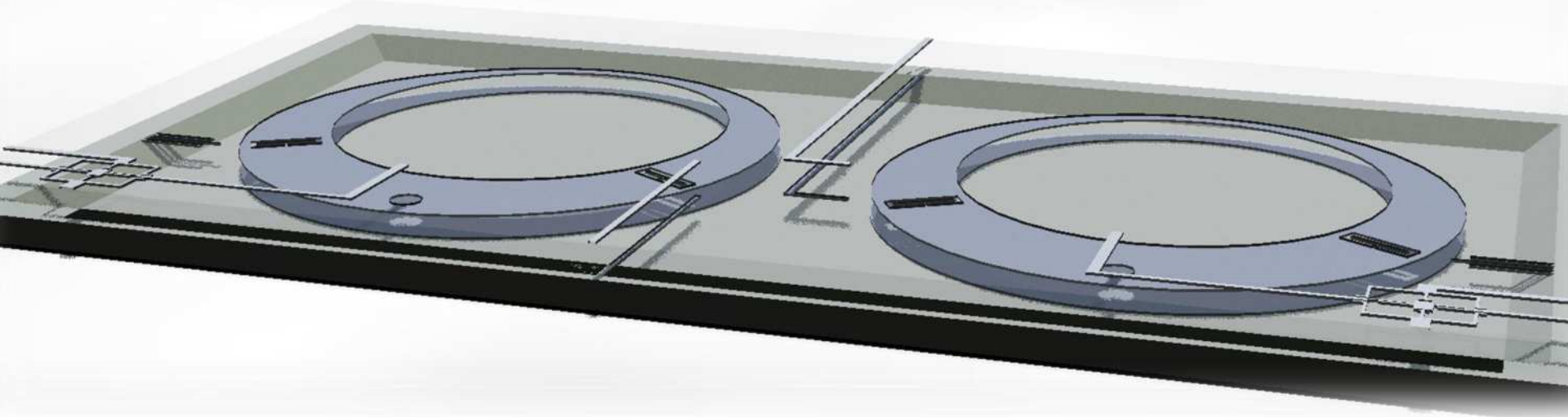


The Co-Axline (MLS 11/14)



WGMRs (MLS 1/13, 2/10/14, 1/15, 2/15)

We dream of:



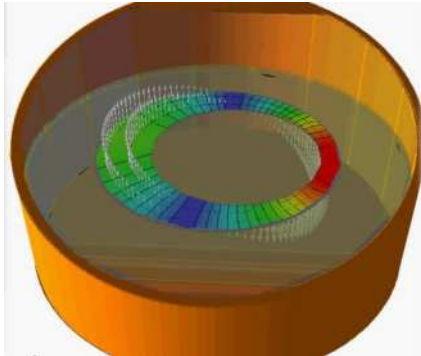
For the moment:

- a. combines qubit & cavities using only wafers
- b. minimal number of moving parts
- c. maximal use of vacuum as dielectric
- d. no seams
- e. readily benefits from material improvements
- f. consistency: both layers from same wafer

In the future:

- a. any qubit (transmon, fluxonium, etc.)
- b. on-chip control lines & amplifiers
- c. layers with specific functionality
- d. full wafer stacking

First generation 3 GHz WGMRs



$$T_1 = 180 \mu\text{s}$$

$$Q_i = 3.4 \cdot 10^6$$

$$T_\phi \geq 1 \text{ ms}$$

$$R_s \leq 250 \text{ n}\Omega$$

$$\tan \delta \leq 10^{-6}$$

APL (2013)



Improved Resonators:

Optical fabrication

7 GHz

$$Q_i = 13\text{M}$$

$$T_1 = 230 \mu\text{s}$$

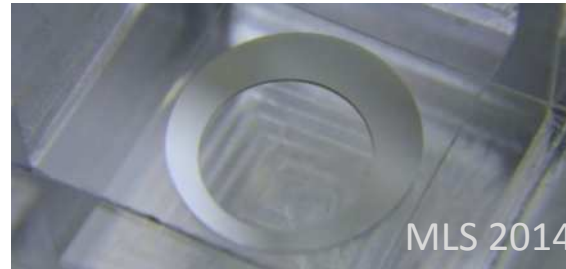
$$R_s < 9 \text{ n}\Omega$$

$$Q_s > 3.7 \cdot 10^5$$



MLS 2015

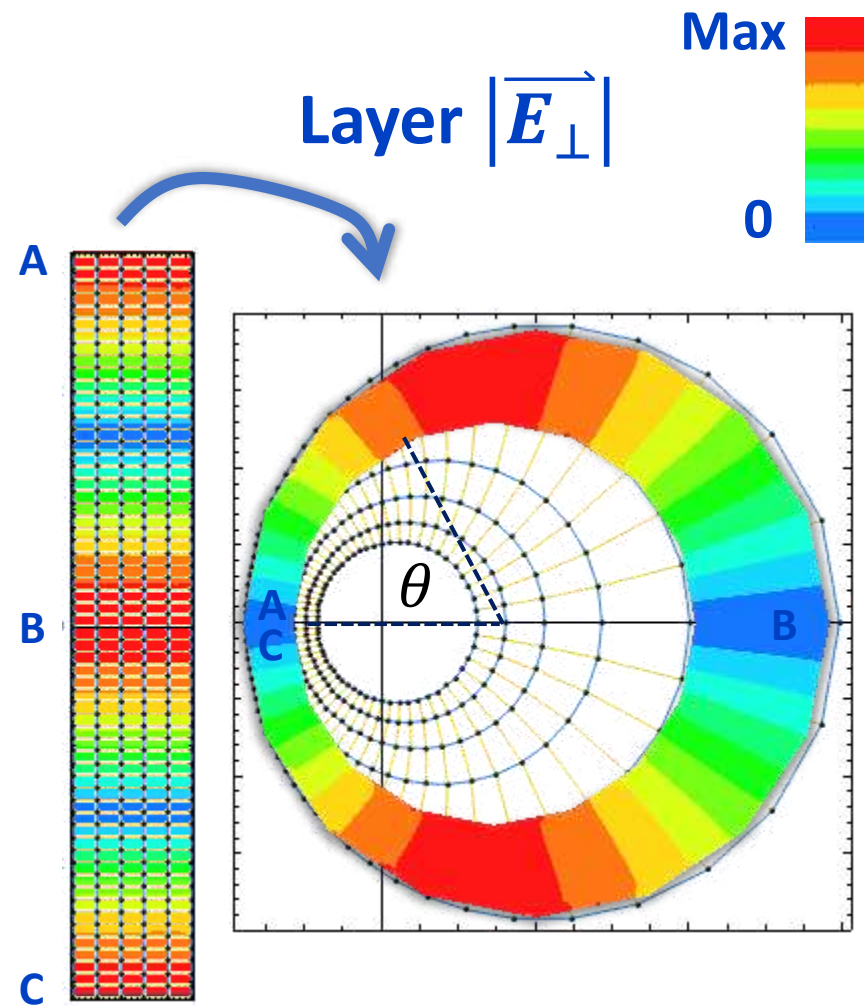
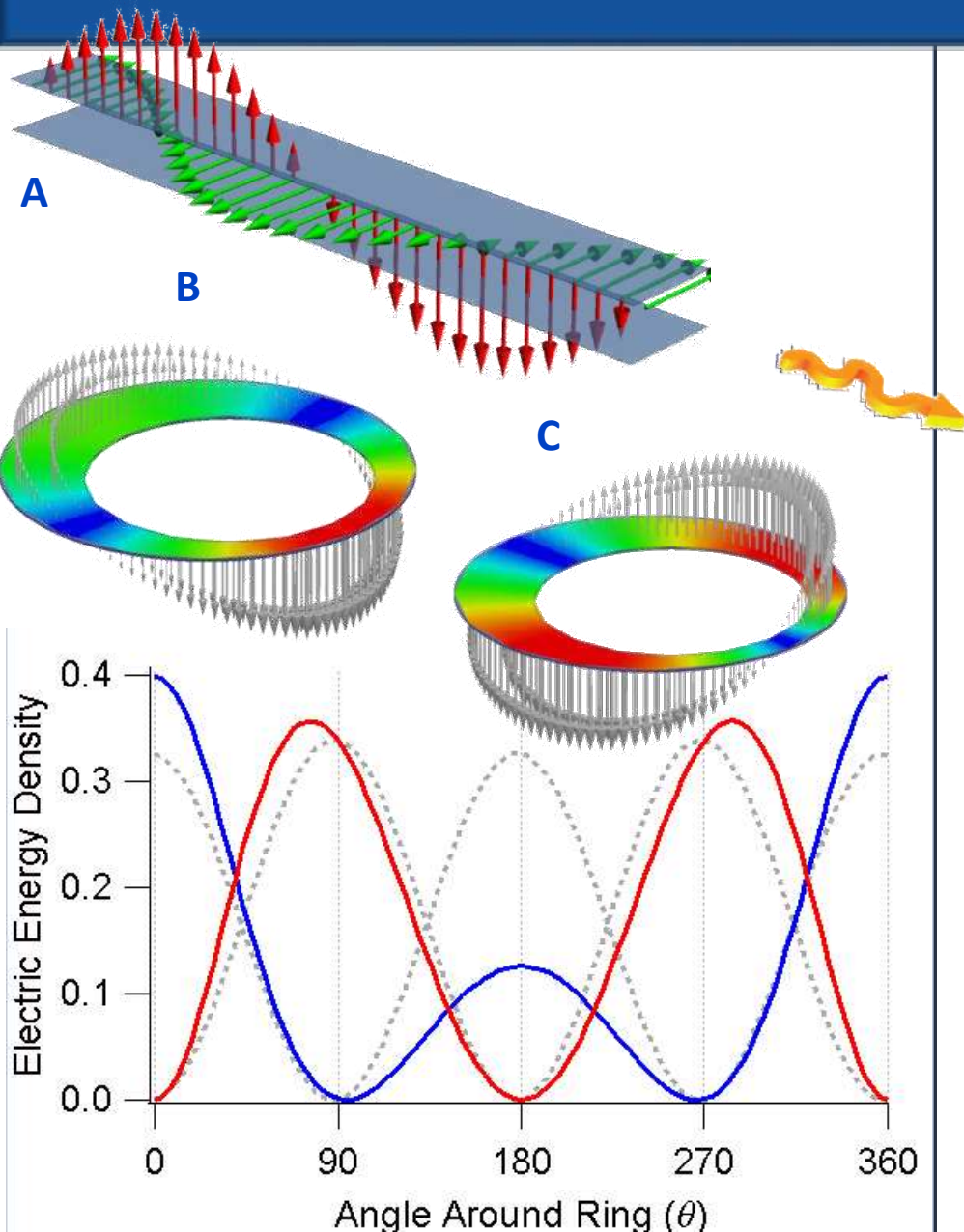
Drawbridge Qubit + Storage



MLS 2014

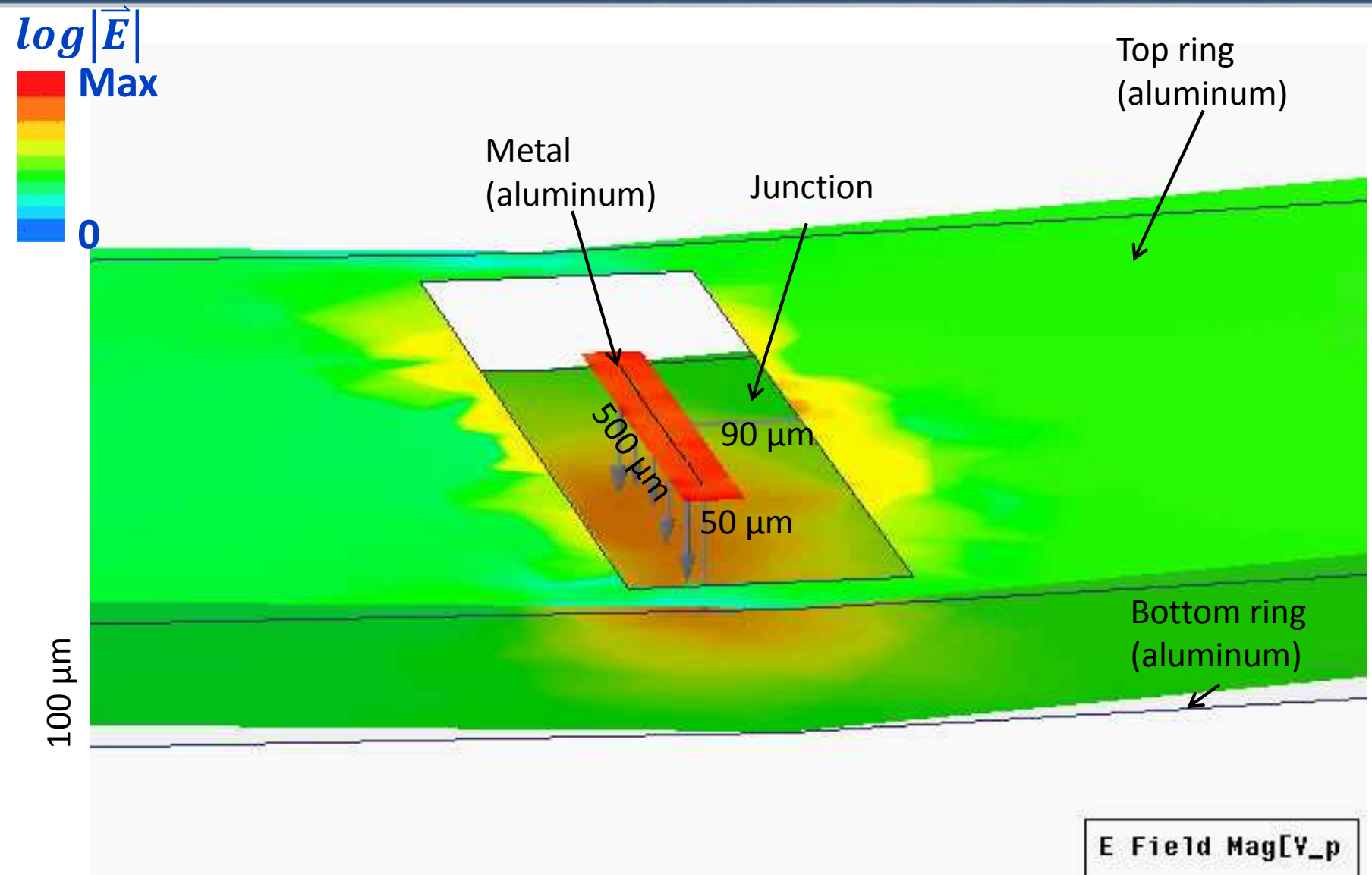
- $T_1 = 70 \mu\text{s}$, $T_2^R = 8 \mu\text{s}$
- $T_{\text{storage}} = 45 \mu\text{s}$
- T_{readout} , chi, etc see later

WGMR Differential Modes (D)

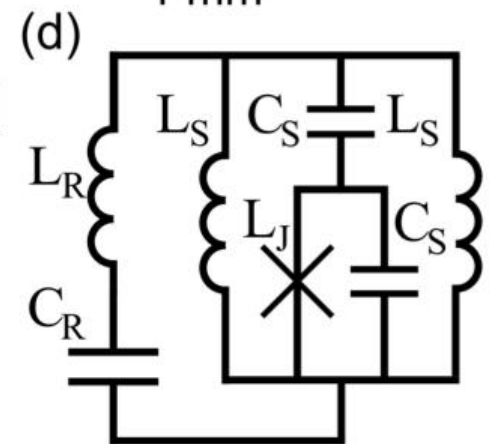
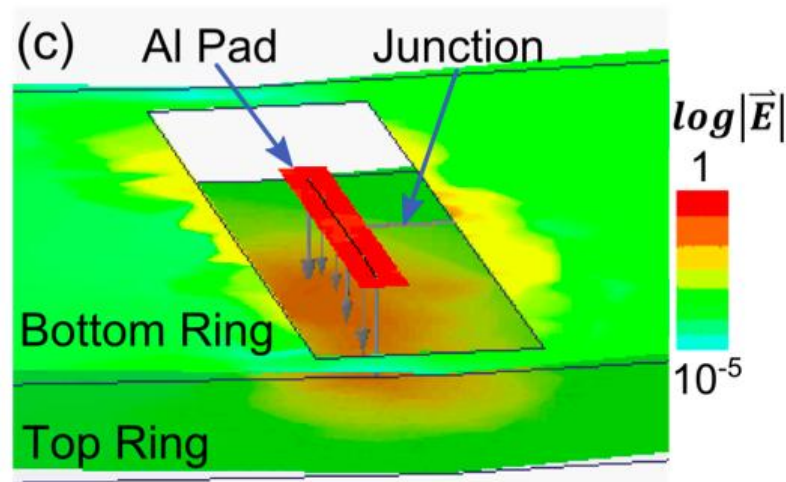
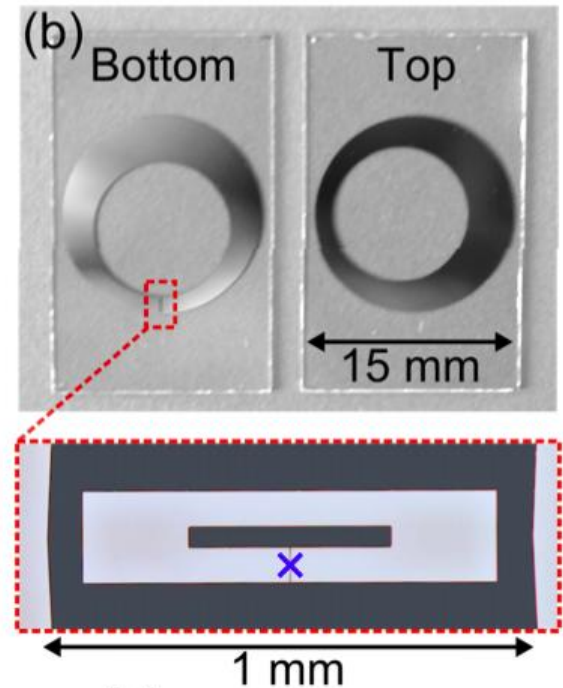
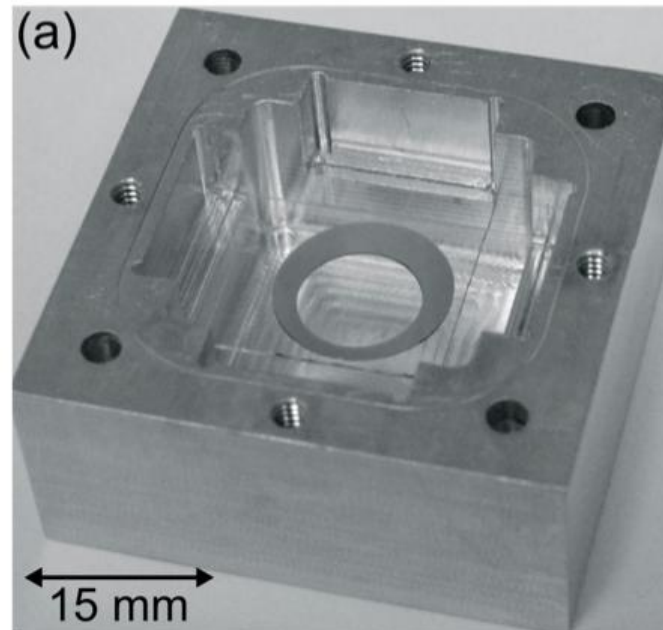
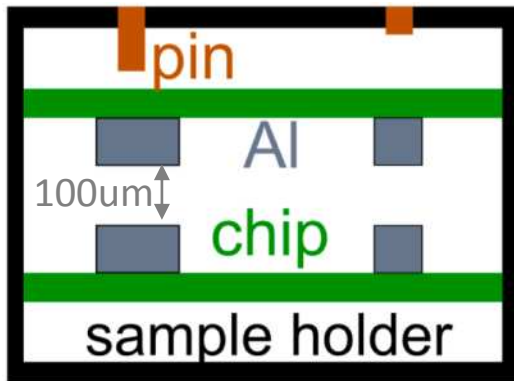


Field expression is analytic and can be obtained by conformal map

Qubit Mode



WGMR Sample

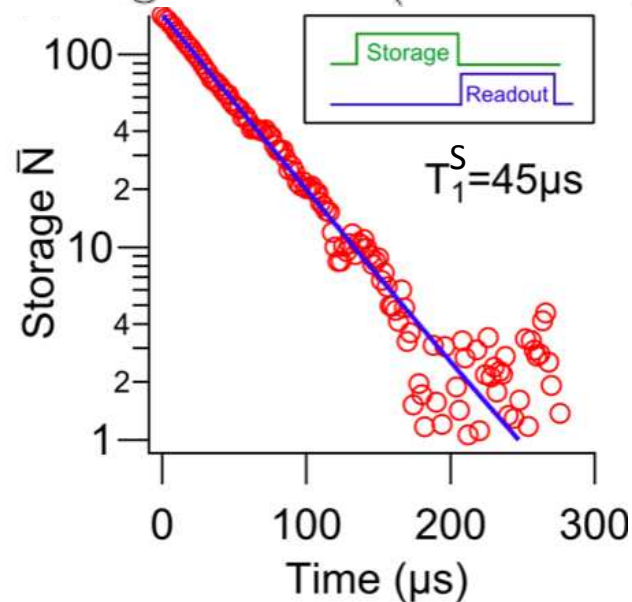


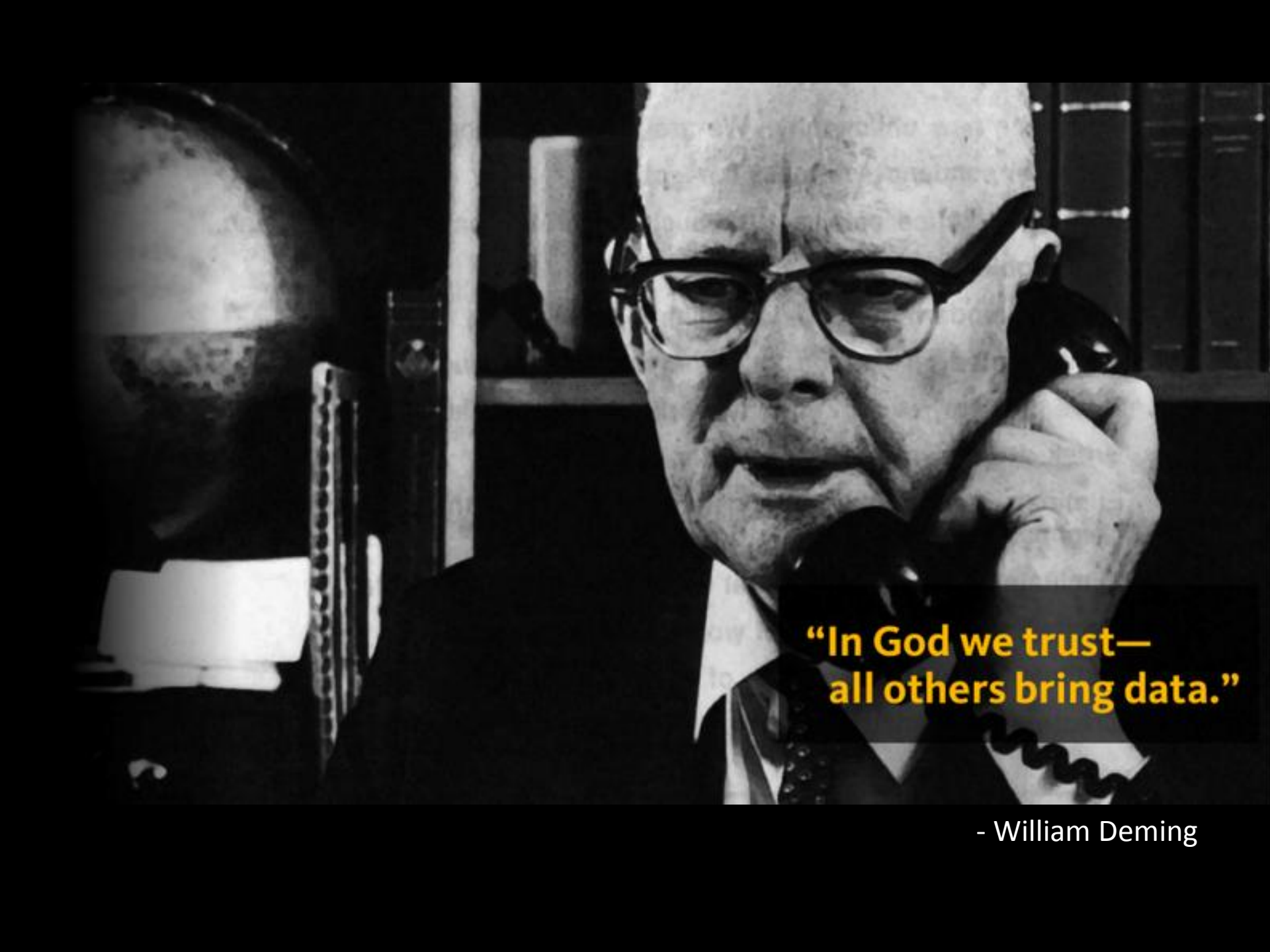
Coherences Summary



Mode	Qubit	Storage	Readout
Frequency (GHz)	4.890	7.070	7.267
T_1 (μs)	70	45	0.42
Q_{total}	2.0×10^6	2.0×10^6	1.8×10^4
$\alpha/2\pi$ (MHz)	300	(5×10^{-5})	(2×10^{-4})
$\chi_q/2\pi$ (MHz)	-	0.23	0.30

Storage lifetime (cross-Kerr)



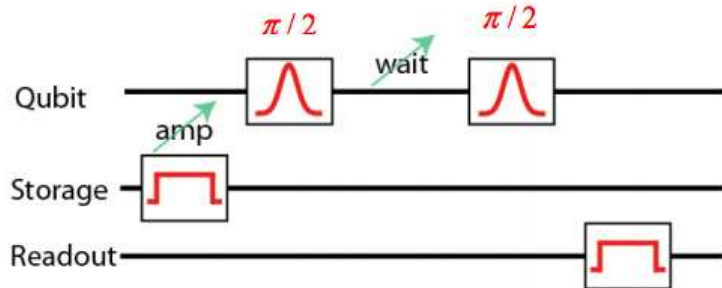
A black and white photograph of an older man with glasses, identified as William Deming, talking on a rotary telephone. He is wearing a suit and tie. The background shows a bookshelf with several books. The lighting is dramatic, with strong highlights and shadows.

**“In God we trust—
all others bring data.”**

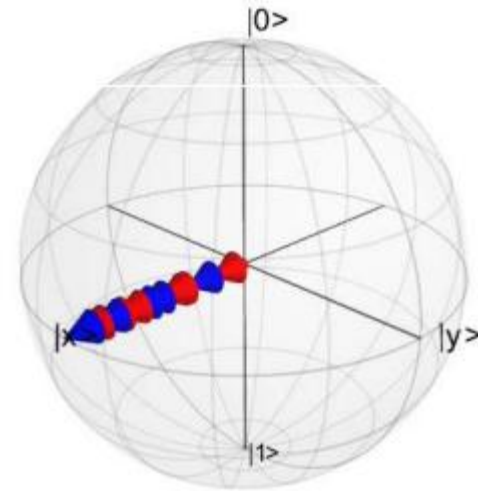
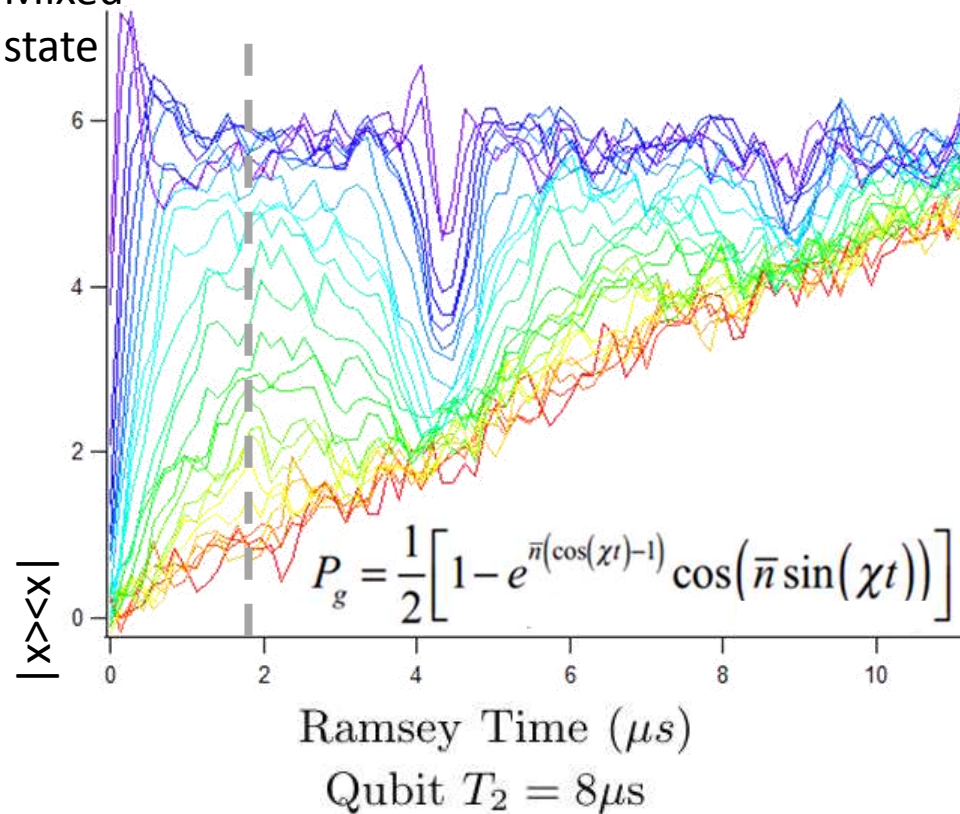
- William Deming

T_1 storage from Parity Measurements

Storage-Qubit Revivals



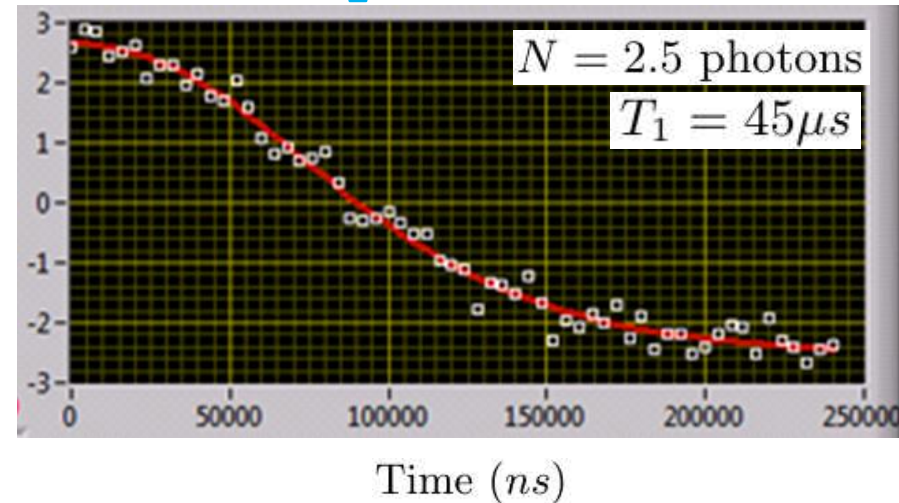
Mixed state



See Zaki & Brian MLS

$$\frac{\pi}{\chi} = 2.25\mu s \quad e^{-\frac{\pi}{\chi T_2}} = .75$$

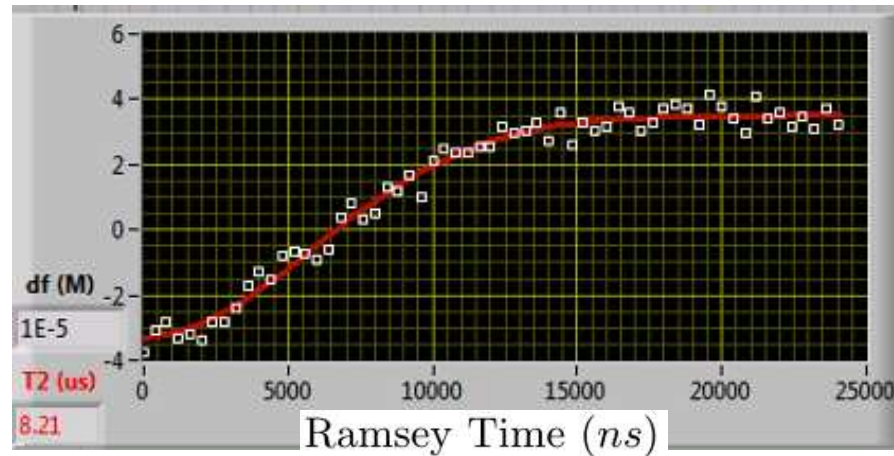
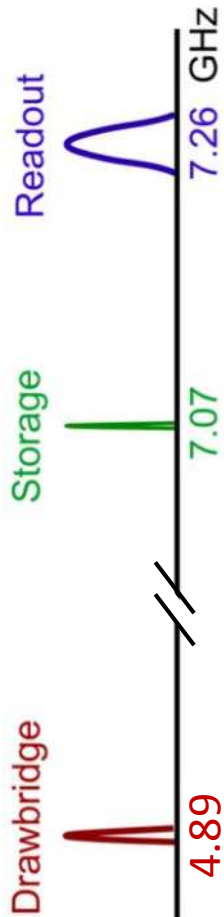
T_1 storage



Parity measurement: Constant Ramsey time = π/χ
Vary time between storage disp. & parity measurement

Qubit T_2

Qubit $T_2 = 8\mu\text{s}$ Gaussian!



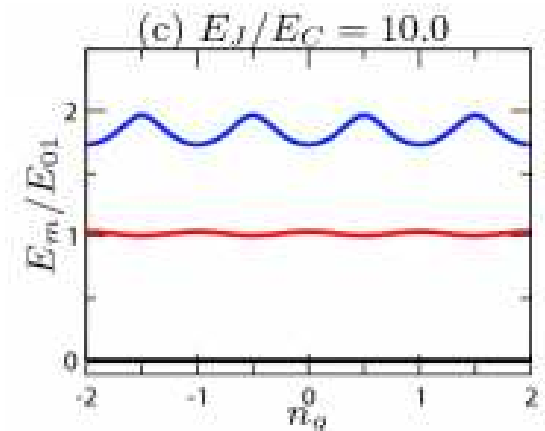
$$\epsilon_m \equiv E_m(n_g = 1/2) - E_m(n_g = 0)$$

$$\text{Dispersion} = -1.30 \times 10^3 \text{ KHz}$$

$$E_j/E_c = 27.5$$

$$E_j = 9.06 \text{ GHz}$$

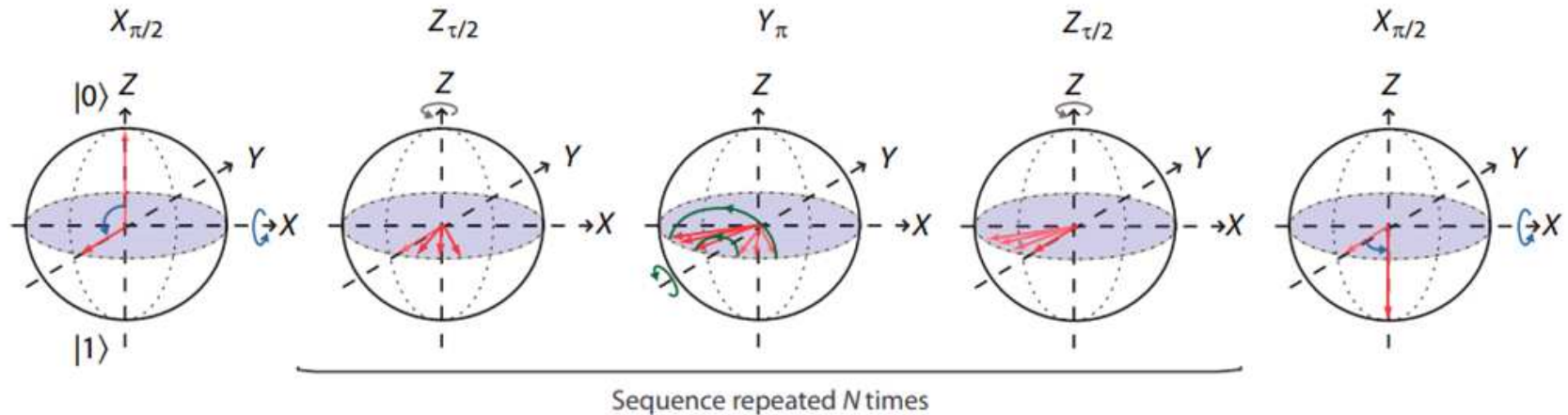
$$T_2/2\pi = 1/(2\pi \cdot 8.2\mu\text{s}) = 19.4 \text{ KHz}$$



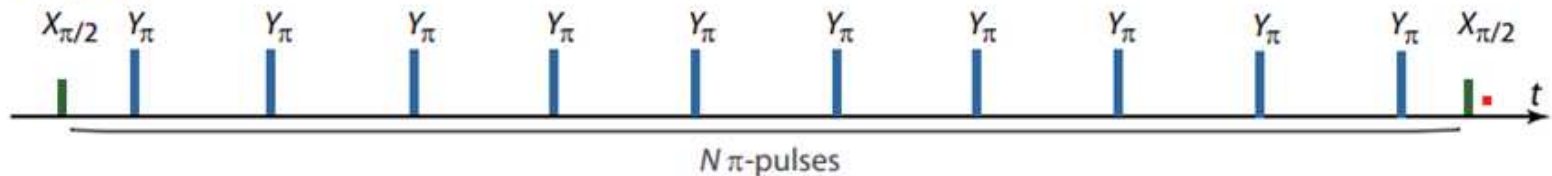
1. What is the noise that limits us?
2. What are the charges doing?

Address Gaussian T_2 by Dynamical Decoupling

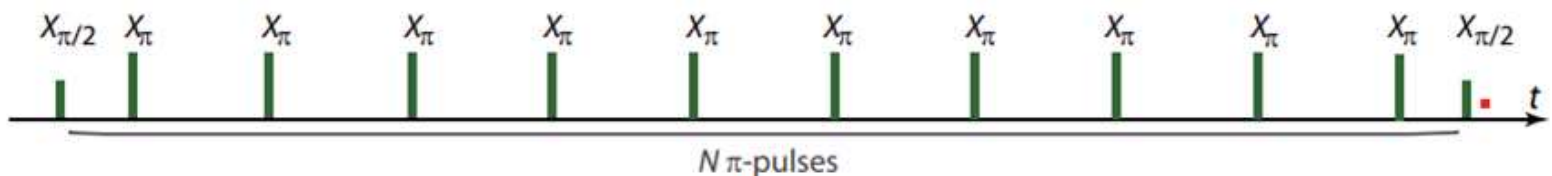
CPMG rotations



CPMG sequence



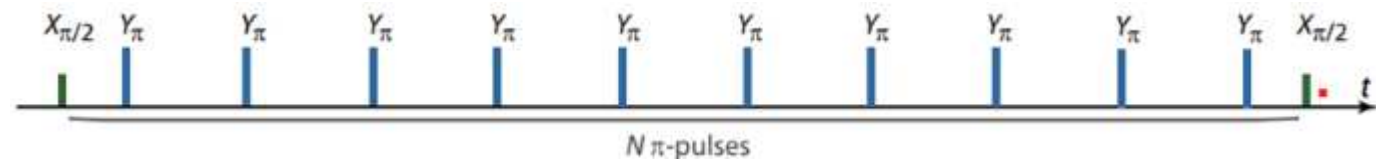
CP sequence



G. De Lange et al. Science 330, 60 (2010)

F. Bylander et al. N.Phys. 7, 567 (2011)

A Filter



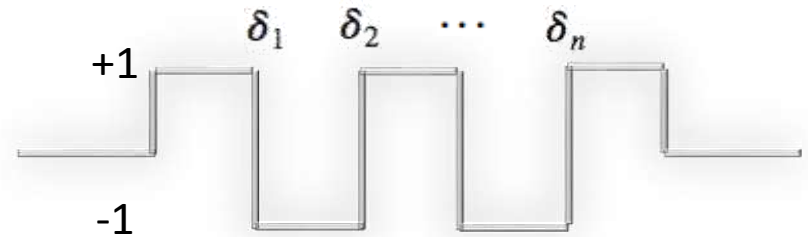
Time evolution:

$$\delta\varphi(t) = (\partial\omega_{01}/\partial\lambda) \int_0^t dt' \delta\lambda(t').$$

$$s_n(t) = \langle \uparrow | D_x^{\text{ef}}(\pi/2)^\dagger R^\dagger \sigma_y^{\text{ef}}(t) R D_x^{\text{ef}}(\pi/2) | \uparrow \rangle$$

$$R = \sigma_y^{\text{ef}}(\delta_n t) \sigma_y^{\text{ef}}(\delta_{n-1} t) \dots \sigma_y^{\text{ef}}(\delta_2 t) \sigma_y^{\text{ef}}(\delta_1 t).$$

$$\delta\varphi(t) = (\partial\omega_{01}/\partial\lambda) \int_0^t dt' \delta\lambda(t') * f_n(t)$$

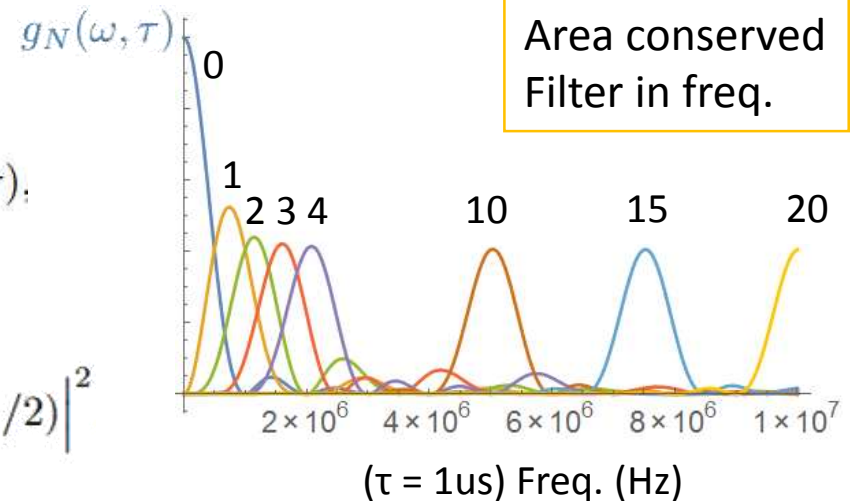


Filtered coherence decay in freq. space:

$$\langle \exp[i \delta\varphi(t)] \rangle \equiv \exp[-\chi_N(t)]$$

$$\chi_N(\tau) = \tau^2 \sum \left(\frac{\partial\omega_{01}}{\partial\lambda} \right)^2 \int_0^\infty d\omega S_\lambda(\omega) g_N(\omega, \tau),$$

$$g_N(\omega, \tau) = \frac{1}{(\omega\tau)^2} \left| 1 + (-1)^{1+N} \exp(i\omega\tau) + 2 \sum_{j=1}^N (-1)^j \exp(i\omega\delta_j\tau) \cos(\omega\tau_\pi/2) \right|^2$$

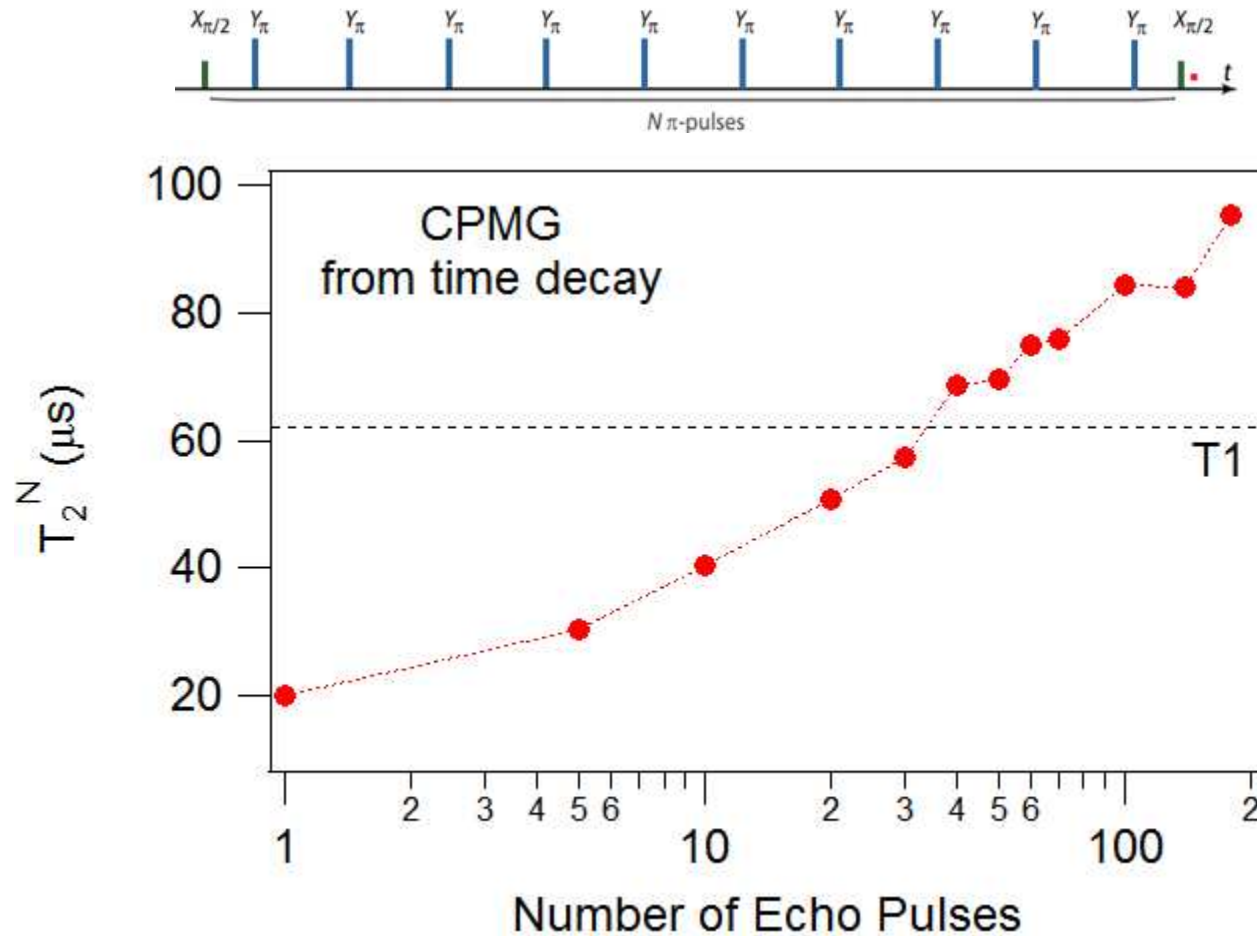


Types of noise:

$$S_\lambda(\omega) = A_\lambda/\omega \longrightarrow \chi_N(\tau) = (\Gamma_\varphi\tau)^2.$$

$$\langle \delta n(t_1) \delta n(t_2) \rangle = \bar{n} e^{-(\kappa/2)|t_1-t_2|} \longrightarrow \langle \exp[i \delta\varphi(t)] \rangle = e^{-\gamma_2 t} \exp \left\{ -4\bar{n}\theta_0^2 \left[\frac{\kappa|t|}{2} - 1 + \exp\left(-\frac{\kappa|t|}{2}\right) \right] \right\}$$

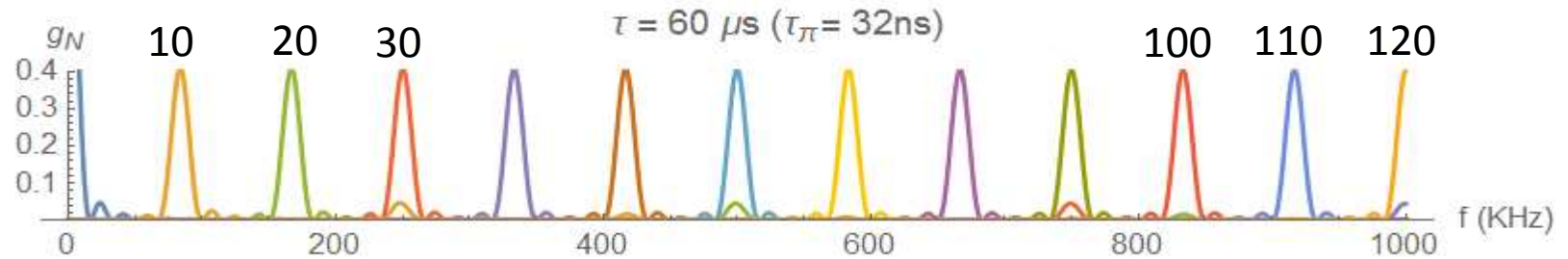
Improvement of T_2 with DD



- Results similar to that of flux qubits but with almost an order of magnitude longer coherences

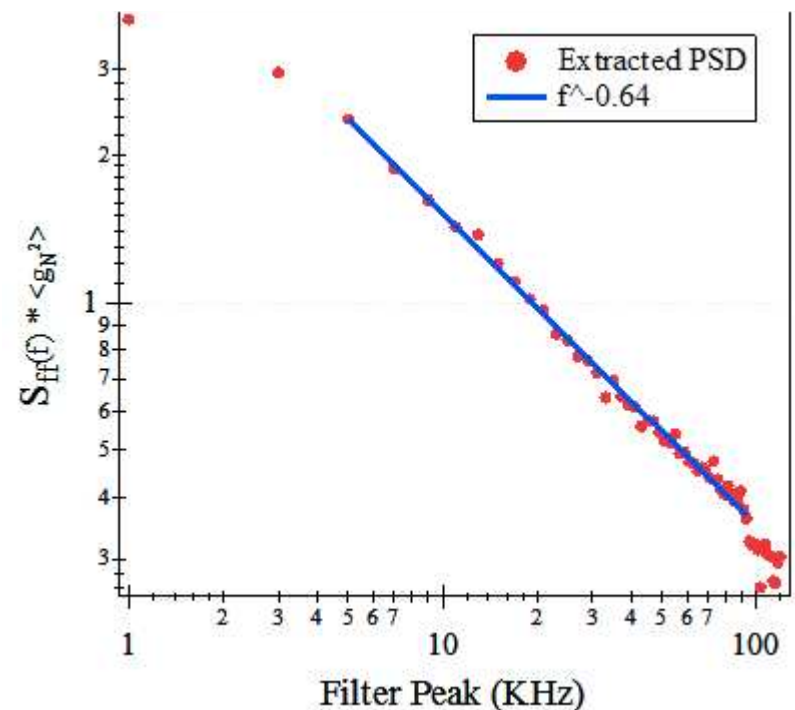
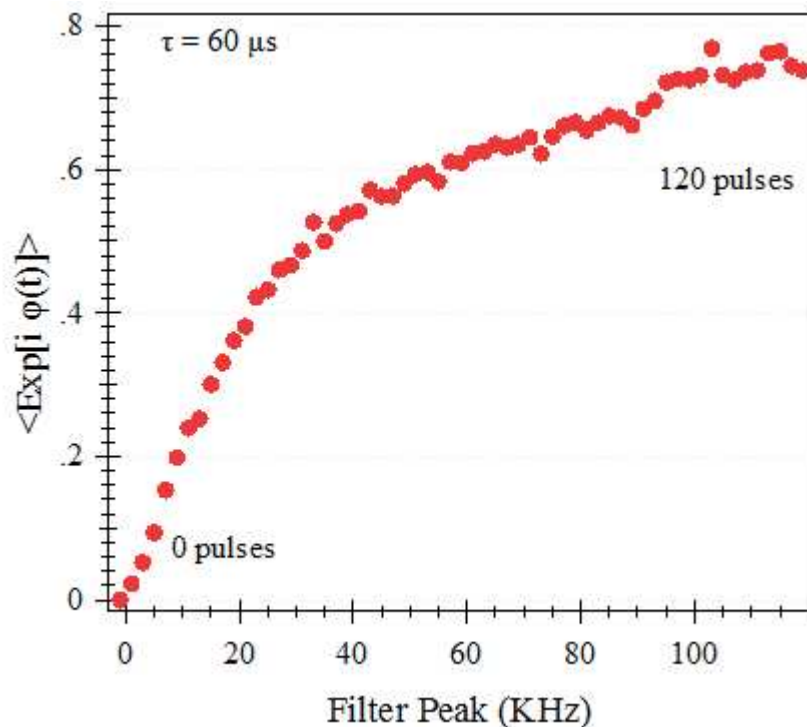
Reconstructing Noise Spectrum

$$g_N(\omega, \tau) = \frac{1}{(\omega\tau)^2} \left| 1 + (-1)^{1+N} \exp(i\omega\tau) + 2 \sum_{j=1}^N (-1)^j \exp(i\omega\delta_j\tau) \cos(\omega\tau_\pi/2) \right|^2,$$



$$\langle \exp[i \delta\varphi(t)] \rangle \equiv \exp[-\chi_N(t)],$$

$$\chi_N(\tau) = \tau^2 \sum_{\lambda} \left(\frac{\partial \omega_{01}}{\partial \lambda} \right)^2 \int_0^\infty d\omega S_{\lambda}(\omega) g_N(\omega, \tau),$$



Quasiparticle

Injection in

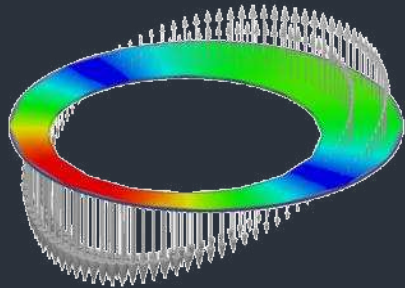
WGMRs



In collaboration with
Uri, Gianluigi, Chen, Nissim

U. Vool and I. Pop, et al. 2014
C. Wang and Y. Gao, et al. 2014

Injection Spectroscopy v. delay

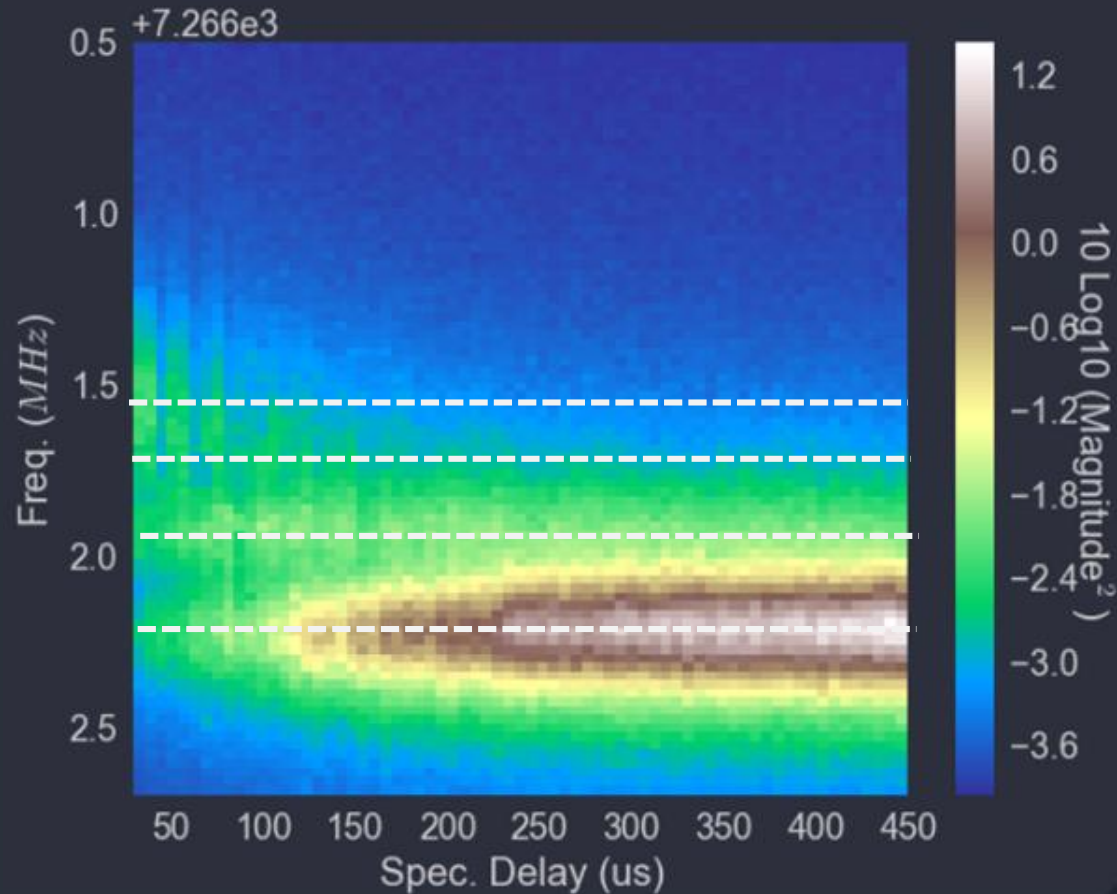


Bright state

$|f\rangle$

$|e\rangle$

Low power g peak



$T_{\text{readout}} = 450 \text{ ns}$
 $k/2\pi = 340 \text{ KHz}$
 $\chi_{\text{qr}} = 260 \text{ KHz}$

Looks very similar for different powers down to a threshold (more on next slides)

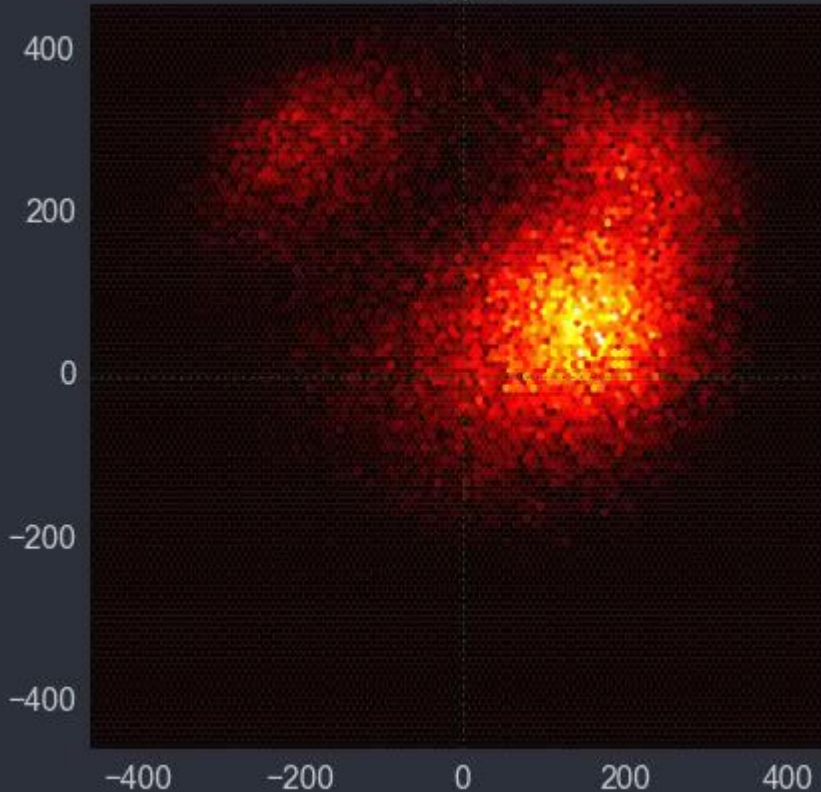
JPC Readout & Resolved Histograms



Delay after injection:

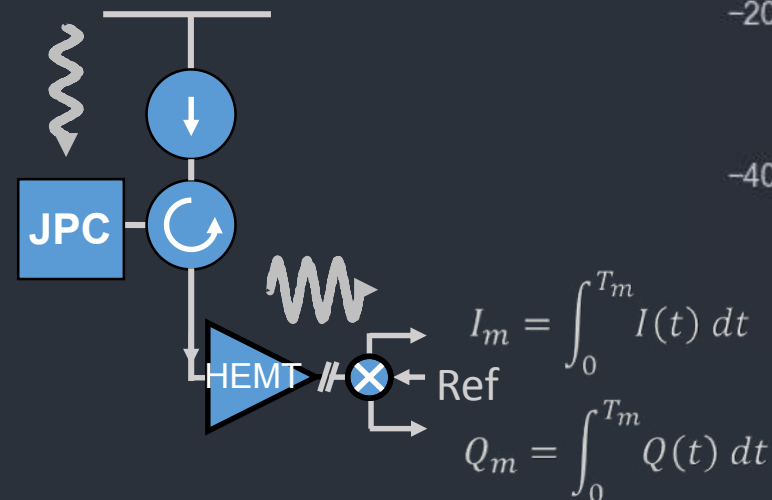
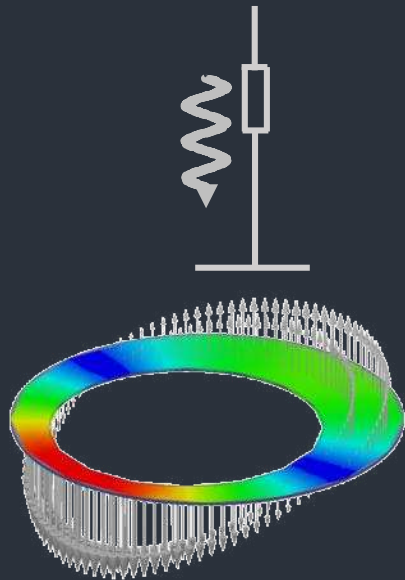
$12\mu\text{s}$

Q Quadrature

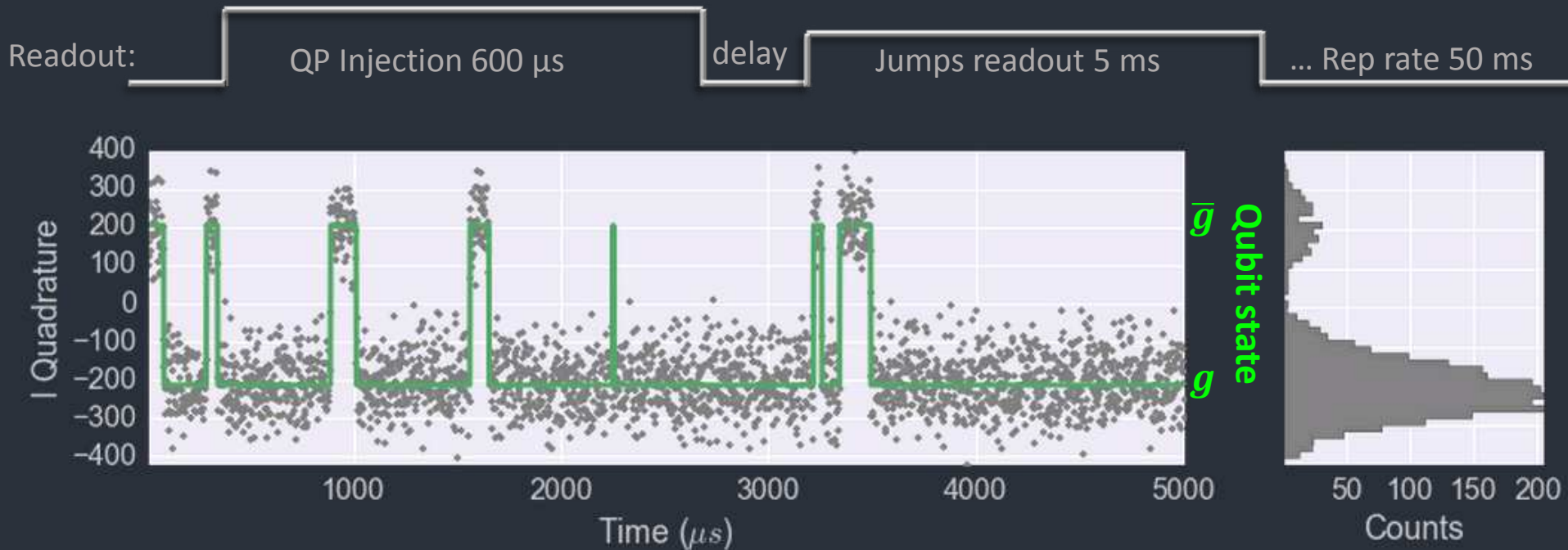


I Quadrature

$2\mu\text{s}$ integration time



Time Domain Analysis (Preliminary)



Three layers to the analysis:

(see Uri's MM rehearsal for more)

1. **Qubit state v. time:** determined from IQ time series using unbiased filter.
2. **Qubit T1 & polarization:** from **qubit state** jump statistics.
3. **QP Dynamics:** x_{qp} , temp. of bath, etc. inferred from **qubit T1 (rates) & polarization**.

A few remarks:

g / not g

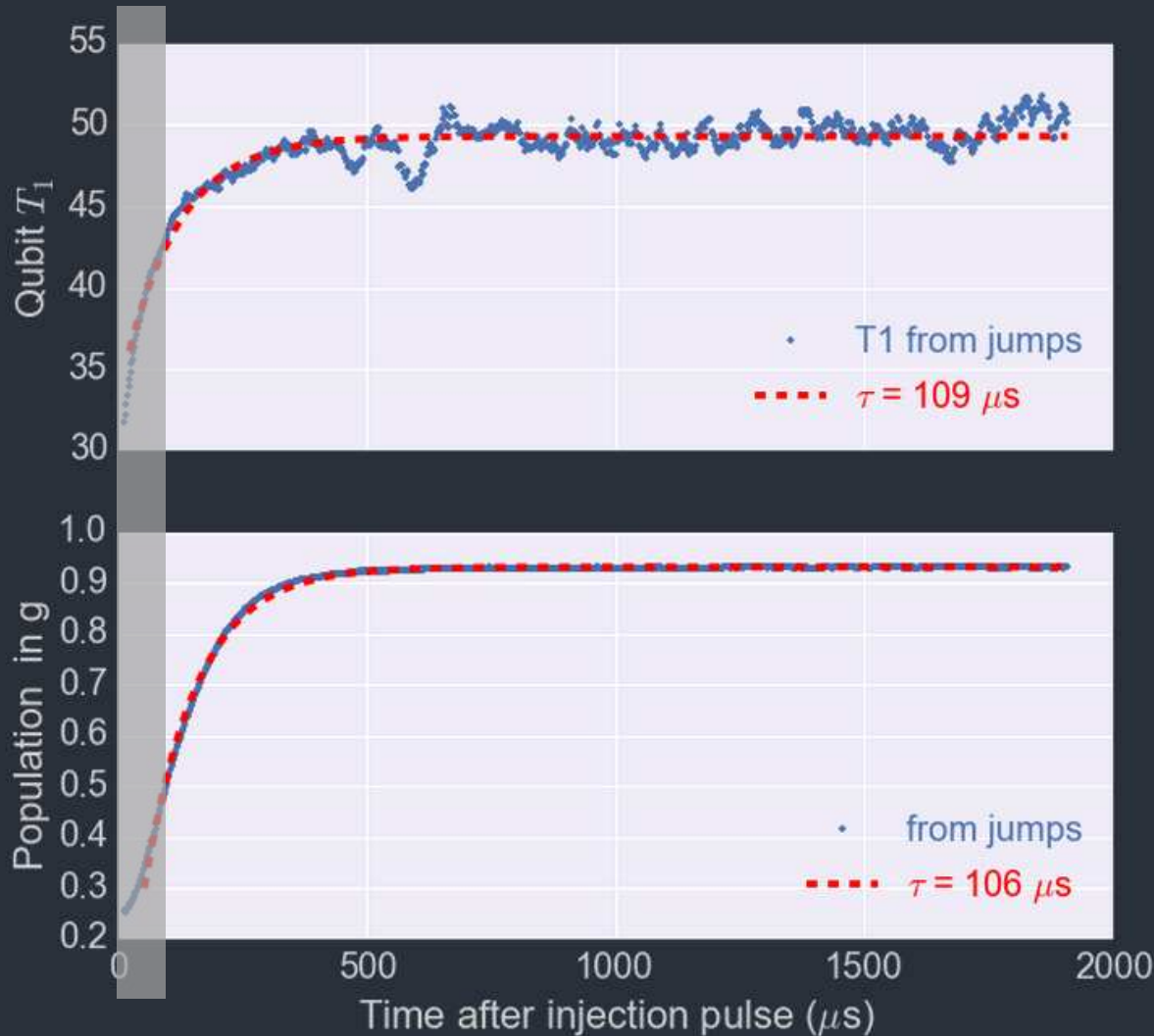
Correlation between measurements

Some biases in filter

But gives us general picture for all states and rates!

2. Qubit T_1 & polarization

from tracking dynamics of the **qubit state**



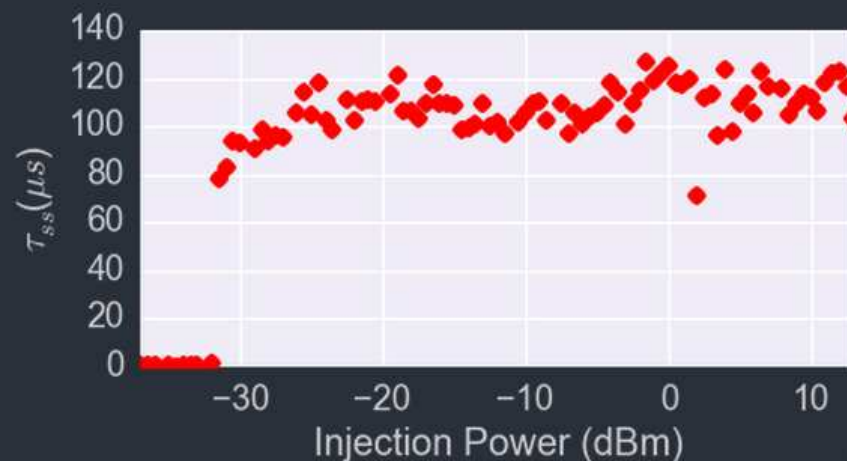
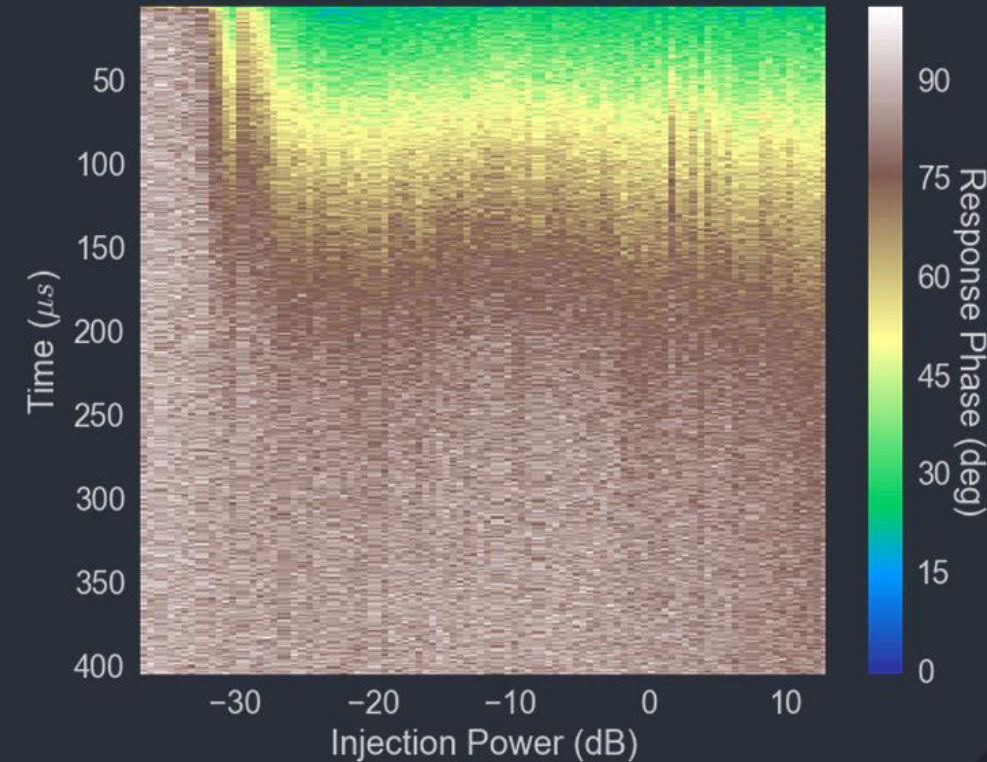
**ignore highly excited region
(more on it in later slides)**

The qubit T_1 is shorter closer to the injection but it recovers on a timescale of $\sim 110 \mu\text{s}$

The population relaxes to steady state on the same time scale.

Highly excited
region

Quasiparticle Injection Threshold Behavior



On left:

qubit population versus time for different injection power

Rates appear to be:

Independent of injection power
Independent of readout power

Quasiparticle equilibration rate to steady state:

Device *	τ_{ss}
Drawbridge Transmon	0.11 ms
Fluxonium	0.13 ms
'Fat' Transmon	0.25 – 2.8 ms
'Skinny' Transmon	5 - 18 ms

A plausible story

(1) Diffusion Γ_D

(2) Vortices Γ_V

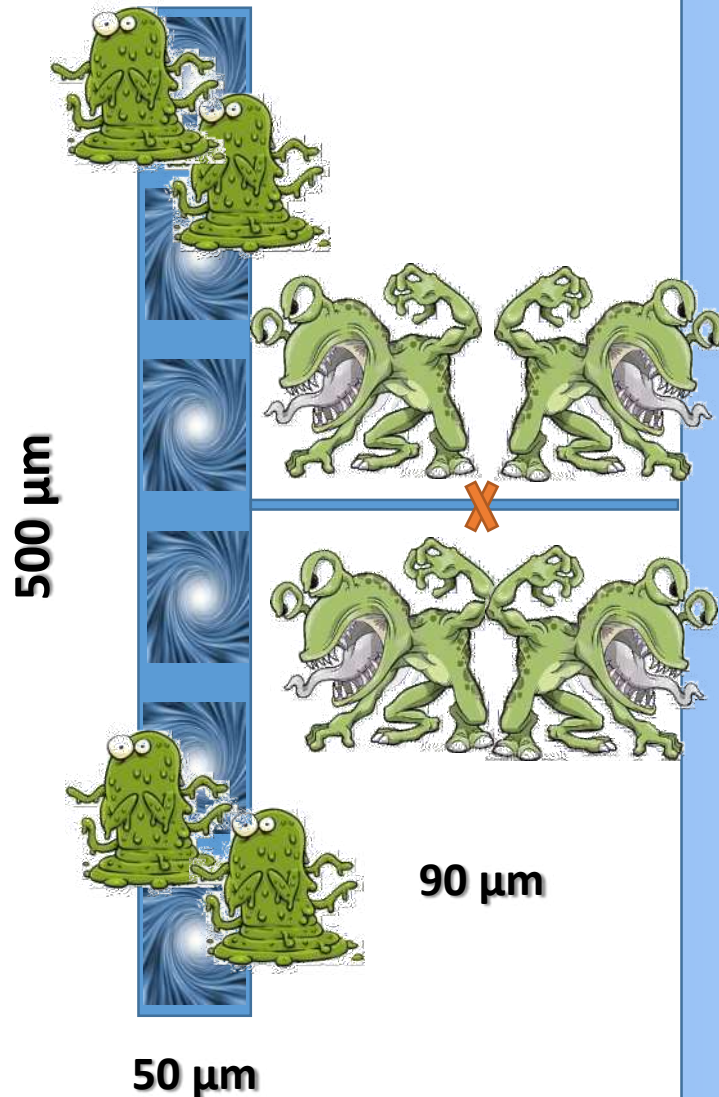
$$\Gamma_{T1} = \Gamma_D + \Gamma_V$$

$$\Gamma_D = \frac{\text{Area} * \text{Length of bridge}}{\text{width} * \text{Diffusion const.}}$$

$$\approx 1\text{ms}$$

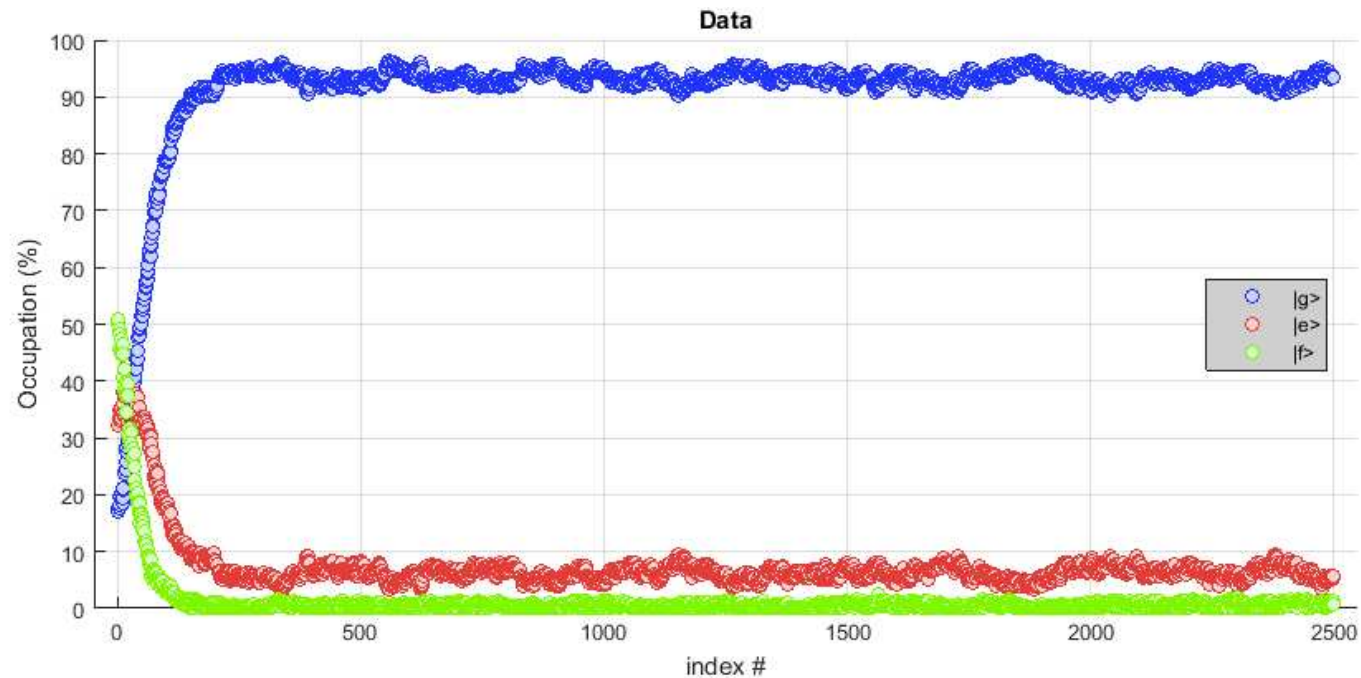
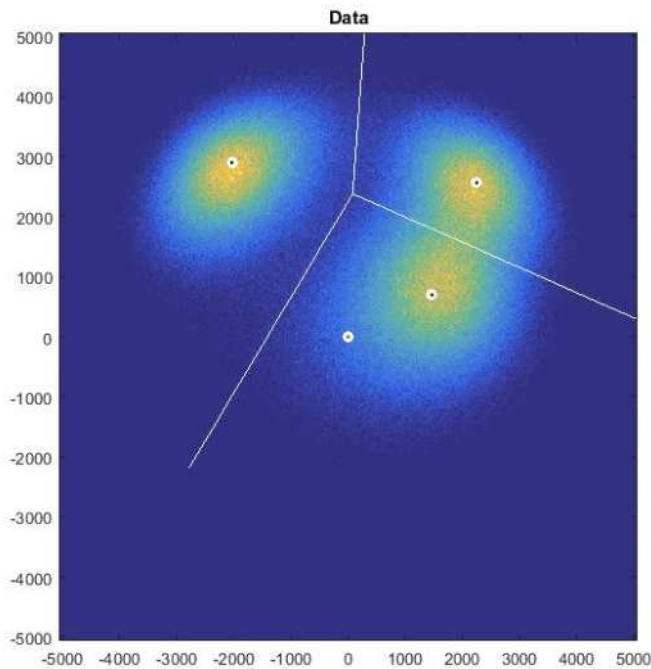
$$\Gamma_V = \frac{N * \text{Trap.power}}{\text{Area}}$$

$$\Rightarrow N \sim 30 \text{ vortices}$$



Some Next Steps

- Extract all rates between g , e , f from IQ time correlations ala **Nissim**



Summary & Conclusions

Proof of concept devices

Achieved

System parameters & photon number splitting regime

Easily engineerable parameters, similar to other implementations

Coherence through dynamical decoupling

Large T_2 s can be obtained if we can get rid of low frequency noise

Quasiparticle dynamics

Fast relaxation (vortices in film?)

Future

Wafer stacking

Multiple qubits

New materials

Fluxonium

